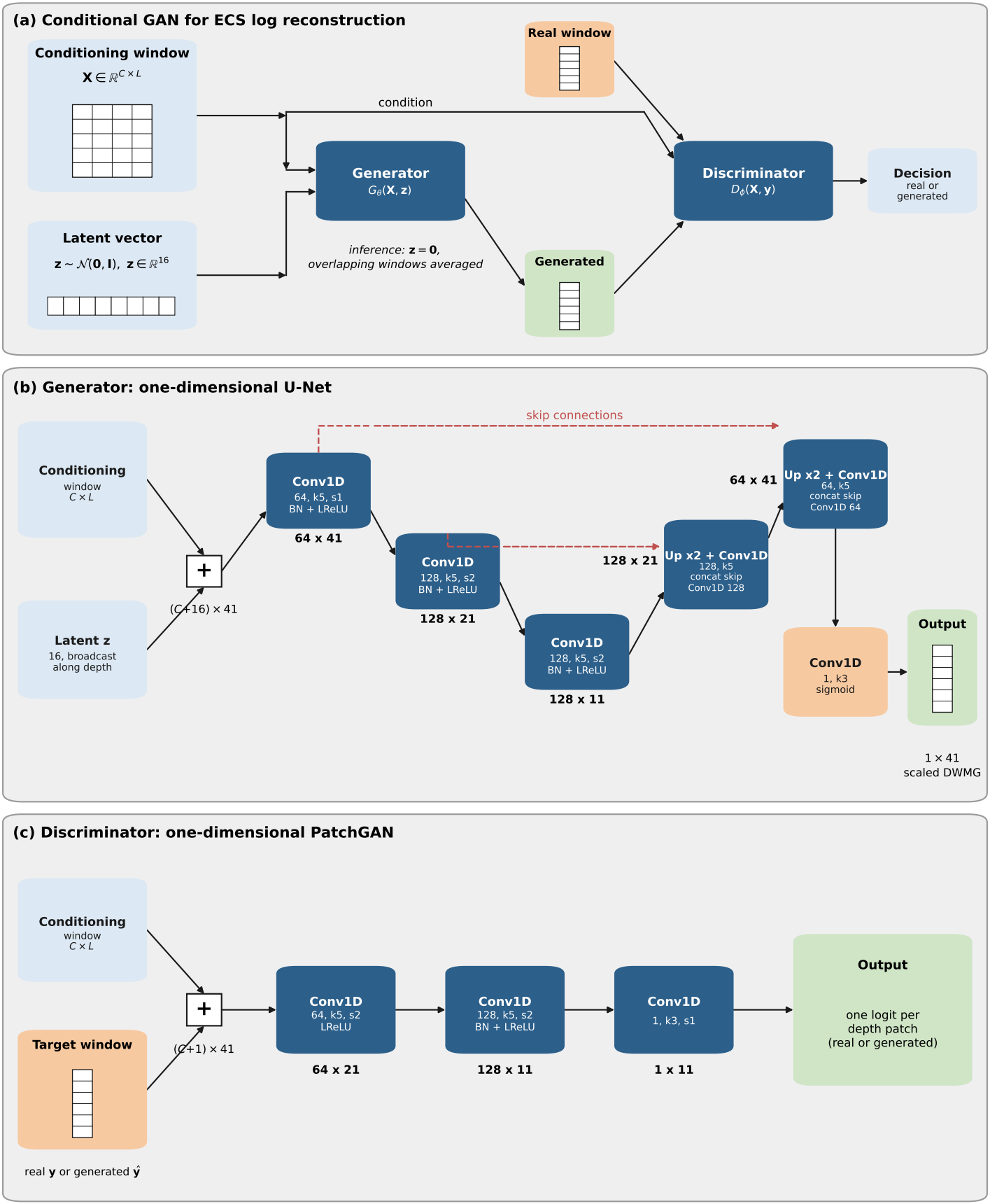
(a) Conditional GAN for ECS log reconstruction
Conditioning window
$\mathbf{X} \in \mathbb{R}^{C \times L}$
Real window
condition
Generator
$G_\theta(\mathbf{X}, \mathbf{z})$
Discriminator
$D_\phi(\mathbf{X}, \mathbf{y})$
Decision
real or generated
Latent vector
$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{z} \in \mathbb{R}^{16}$
inference: $\mathbf{z} = \mathbf{0}$,
overlapping windows averaged
Generated
(b) Generator: one-dimensional U-Net
skip connections
Conditioning
window
$C \times L$
Latent z
16, broadcast
along depth
(C+16) × 41
Conv1D
64, k5, s1
BN + LReLU
64 x 41
Conv1D
128, k5, s2
BN + LReLU
128 x 21
Conv1D
128, k5, s2
BN + LReLU
128 x 11
Up x2 + Conv1D
128, k5
concat skip
Conv1D 128
128 x 21
Up x2 + Conv1D
64, k5
concat skip
Conv1D 64
64 x 41
Conv1D
1, k3
sigmoid
Output
1 × 41
scaled DWMG
(c) Discriminator: one-dimensional PatchGAN
Conditioning
window
$C \times L$
Target window
real $\mathbf{y}$ or generated $\hat{\mathbf{y}}$
(C+1) × 41
Conv1D
64, k5, s2
LReLU
64 x 21
Conv1D
128, k5, s2
BN + LReLU
128 x 11
Conv1D
1, k3, s1
1 x 11
Output
one logit per
depth patch
(real or generated)